Is The Brain A Statistical Organ? The Free Energy Principle & Active Inference | Karl Friston

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[Music] thank you I've been reading your work for many years and you are someone who's managed to incorporate so many different fields together at this point your free energy principle can be discussed in fields of mathematics physics biology it seems you've managed to cross all these borders and I think it's perfect for this show one thing I want to start off with is when you think of the topic of the mind-body problem you think of Rene Descartes and he often said I think therefore I am and you've now become synonymous with I am therefore I think I think let's start off with that because I think it's beautifully poetic considering this podcast is called Mind Body Solution yes I I do like that um homage to uh Descartes um so the the cheekiness of the reversal uh is based upon the free energy principle um read as a physicist and effectively this is a description a principle that will be applied to anything that exists um and the maths tells you that if something exists in a certain sense and then it can be read or interpreted as making inferences so if you allow me to equate inferring with thinking that means to exist is to infer so if I am therefore I think so that's the Genesis it's one of those um done that it's um strange inversions yes you know uh speaking to the slightly sort of deflationary aspect of the of the free energy principle you know it's slightly tautological aspect I think at that point it would be a perfect opportunity to let's play with definitions let's talk about three energy principle a Bayesian brain and perhaps active inference let's play around with these definitions before we move to a formalism of the free energy principle right um so they're all very closely related uh so defining the free energy principle first of all what is it it's just a principle of the kind that you would remember at school in physics like Hamilton's principle of least action um and as such it is effectively a mathematical method it's a method or a tool that you would apply to a particular situation and when applied to things like you and me or in fact many other animate things and biotic things um one gets an interpretation of the behavior of such things in terms of active inference um the inference part is just a statement of sense making and representation on the inside of the thing uh where the thing in question can be a person or a particle it could be you or me um and that sense making speaks to making sense representing explaining inferring the causes of sensory Impressions from everything else from the outside world um and the best or the most if you like inclusive way of understanding that kind of inference and sense making when applied to the brain would be the Bayesian brain hypothesis so the Bayesian brain hypothesis is the notion that we can understand our sense making and to a certain extent with one commits to neuronal process there is the underlying neurophysiology neuronal Dynamics in terms of Bayesian belief updating basically changing our physical by physical States via neurodynamics to represent probabilistic beliefs bayesians of personal beliefs probabilistic beliefs about the causes of Sensations so that's inference inference um Loosely described under the Bayesian brain active inference is the simple move that you are acknowledge that the sensations upon which you're basing your inference are actively sampled you are in charge of the data that is subtending your inference and that sort of closes the loop between you as an inference machine a statistical organ or your brain at least as a statistical organ and and the world supplying that sensory evidence of those Sensations and that story and that sort of closing the loop uh an action perception kind of cycle a description of sentient behavior um appealing to the Notions of active uh sensing and active learning um is one way of reading what comes out of the free energy principle so that's in my mind how those things uh oh let's see how you triangulate amongst those three Notions and when we think about I mean obviously as you said they're all very linked and when you think about the free energy principle the formalism of it do you think of energy minus entropy and then you obviously go into different aspects surprise Divergence then you discuss obviously your internal States hidden States action perception talk to us about how this formula actually works and how you do because a lot of people have noticed when they question you about this Theory everyone comes from a specific field so if someone often tends to attack you from a psychological perspective someone tends to talk to you from a biological perspective how do you navigate all these territories and I think with that I'd like to say that you're one of the few people that I've noticed who seems to be a genuine polymath in that regard because you're always able to answer these questions succinctly with a lot of definitions and just beautifully how do you manage to do this I have a lot of naughty and inquisitive students so you get used to answering questions uh after a while um yeah that is again very very gracious of you to say that um but it's not I I don't think it's um my wisdom um I think that ability to jump from um different disciplines or just take different perspective or views on what is probably the most important thing to all of us which is how do we work and why are we here um is the gift of things like the free energy principle and the free energy principle I'm not claiming that that you know the 300 is just the current formulation of um Millennia of thinking and physics and psychology and philosophy um and probably nearly every other field including behaviorism economics and the like so the whole point of the preemption principle is that it is um there to try and unify and to try and Encompass and thereby endorse all the various perspectives you can take upon centered Behavior or Choice uh the uh the imperatives and the right our decisions um you know right through where do I look next in the next few hundred milliseconds through to you know which school should I send my children to and so the the eclectic aspect that is brought to the table by the free energy principle is just due to its fundamental nature it's it's a first principle account and by definition if it is a first principle account if it is applicable that it should be applicable widely to various phenomena red or looked at through the lens of an economist or a psychologist or a physicist or or a philosopher um that that it's interesting to say I talk about things like surprised information Theory and entropy and um intrinsic value you didn't actually say that but I do talk about that affordances and the the whole charm of having a first principal account that is sufficiently simple to write down mathematically is that you can see that the same mathematical quantities mean different things to different people but they're all the same thing so perhaps you know just to give you a couple of examples of that underneath all of this is just basically a description of the states that I exist in those states that are characteristic of me so how would I express that mathematically well I could just write down the uh a function of any particular state that reflected how characteristic it was of me and that would be most commonly written down as a log probability of me being in this state now any log probability is basically a potential and if it's a potential you can now read it in by analogy to thermodynamics um that log probability of being in the particular state if I am me can also be read as a statistician and you can call that model evidence or in statistics marginal likelihood if your information Theory would be called self-information and the average of self-imation is entropy um if you were an economist it would basically be the states that I aspire to it would be description of my attracting set written as if you were a theorist dealing with random dynamical systems so now this same quantity now becomes an expression of um utility or value if you were ET James you would call it a constraint on the imperative to maximize the entropy and so on and so forth so the key thing is that everybody's talking about exactly the same thing it's just that they've got different words from it so the you know the joy of um sort of seeing that um Central quantity under the hood being able to make sense to lots of different people just is the um an expression of the expiratory scope of this kind of first principles account so everybody's right they've just been using different words I I think you're absolutely right I think when people try to develop any Theory or hypothesis it's always very important to sort of realize that we don't exist in a vacuum there's there's so many different fields of study that are studying the same things but just defining them differently um discussing them very differently but they all come together at some point and any good theory needs to be able to explain these various expertises as well there needs to be some sort of a full explanatory version of all of these and they shouldn't be a gap and you manage to bring these all together when you talk about hierarchical Bayesian inference it plays a very Central role in your work and I remember when I was reading about this and doing computational Psychiatry as part of the blocks during my masters I remember coming across your work and thinking this is so beautiful this is the way you've managed to sort of incorporate this into Psychopathology is even more intriguing because I remember the first time I read about it and you spoke about posterior and priors and the way it will dictate how someone experiences hallucinations delusions and just general differences in their field of vision Etc how did that come to you at some point at what point did you manage to realize that this Theory as a very active role to play in mental health all right well the answer is another strange inversion I'm afraid I started with me um I started as a psychiatrist um and a what used to be called a sort of mathematical psychologist or what I wanted to be a mathematical psychologist and I think a clinical version of that is exactly as you say a computational psychiatrist that phrase introduced by my friends uh Reed Montague and Peter Diane um has only been around for about half a decade but now we're all computational psychiatrists so as a young man I aspire to be a computational psychiatrist so the free energy principle is the denouement of um that formalization but I think it's a really important point you bring to the table there um and and I think there are two moves you um have to make to to appreciate the potential applicability um of things like the Bayesian brain of the three energy principle to to the suffering of patients that you might see in the outpatients the first move is to have a physics of beliefs a physics of inference a physics of sense making if you like a physics of basic sentience uh that is inactive so it has to be sentient Behavior so it's all about choices and decisions and how I move and indeed um how I control my autonomic reflexes you know respond to my gut feelings and exert from some personal control over my physiology and internal learn um so that's the first thing that you have to have you have to have a physics of sentence you have to have a mathematics that connects the uh the foundations of our

Dynamics in a physical world to the world of beliefs and influence and sense making and representation in this inactive sense so that's the first thing then once you've done that you can then say well okay under what circumstances or what would it look like if this inference went awry was in some sense not fit for purpose of the particular world or Eco Niche or environmental interpersonal relationship that this person was having to infer and to learn and to actively navigate to deploy her active imprints for example um and then you get I think to computational Psychiatry but that does depend upon um committing to a view of all psychopathology as intrinsically some kind of false um but it's a very natural view that if you just think about hallucinations and delusions for example I mean what are they they are effectively inferring something is there when it's not so if you were a statistician you say oh yes that's a type 1 error that's just bad influence because you've got something wrong with your standard error but of course you know that there's some direct corresponds to the Precision in uh say predictive coding variants of the basic brain hypothesis um which itself then relates to synaptic gain and neuromodulation and certain the actual certain drugs um similarly all neglect syndromes um or um in the old days would have been called dissensitive or hysterical syndromes you know again read as false inference which simply mean I'm inferring something is not there when it is it's a type 2 error um and you if you think about all the different disorders and ways of suffering uh that people present with in Psychiatry and indeed quite a lot of Neurology as well there can all be casts as some kind of false inference you know um dysmorphophobias um say anorexia nervosa having you know false inferences about my body and even simple things like uh parking that simple but awful things um but uh you know simple in the sense the more motor and neurologically oriented um conditions like Parkinson's disease you could think of as a failure to infer and to realize my inferred intentions that I am now about to move you have made tumors first of all we've linked physics to belief updating and inference and then we've taken um Psychiatry as instances of a kind of aberrance or a kind of failure of that influence and belief updating process but I repeat put it in an inactive context so you know you couldn't explain Parkinson's disease unless you talked about um Banning as inference you know treating movement and acting engagement with the world as the realization of inferred fans things that I am likely to do next in in this moment or indeed again choosing which school to send my children to I still remember there was one specific line that I quoted from one of your papers I can't remember the name of the paper but you actually spoke about that phenomenon way someone who's acutely psychotic schizophrenic when you look at these patients and you show them visual illusions you're able they're able for some reason to see the reality of the situation whereas we fall for this trick so in those cases people often say that they're seeing the real reality the radical truth and we're not talk to me about this phenomenon because I think it's quite intriguing that they don't fall for it but you with the free energy principle with your theory you're able to explain this it's a fascinating area so that um there are rare instances where um Psychopathology or any pathology um can actually improve certain performance um of course one has to qualify what one means by improved performance but the particular example you're talking about I think is absolutely fascinating and it's a resistance um characteristically shown with certain people either people on the Spectrum or people with schizophrenia uh or schizo typo um approaches to life um that renders them uniquely resistant to the kind of illusions that we all experience um and it's just worth noting well what's an illusion well it's a false inference but why is it false it's perfectly good from the point of view of the person having the the inference and having the uh presumed percept and qualitative experience uh however the experimenter you know the naughty experimenter who's deliberately after the stimulus in a way which is so unlikely that they can now say no no this is a false percept because I know how it was caused and the subject I for example have completely the wrong idea the wrong influence about how it was caused so it is a wonderful tool for people in psychophysics to actually test how our inferences can fail in relation to the actual causes of our Sensations but to put that another way around what it does it's a beautiful Pro of the prior beliefs that we bring to the table so that you know going back to the notion of the Bayesian brain read simply that means that we have um at hand a number of hypotheses that can explain the causes of this pattern of sensory or for example visual information and we generate these hypotheses actively say predictions in predictive coding and then we look at the adequacy of these Protections in um explaining the sensory data so effectively little scientists we are testing hypotheses against sensory data and if they're not good at hypotheses we will reject them and um and revise our hypotheses and this is the basic belief updating that we're referring to previously but they um that um that adjudication and that updating rests upon your prior beliefs before you see some sensory data um relative to the Precision of those data that are going to move your prior beliefs to make them into post-drivilees after seeing the data so we come back again to this notion uh and a key notion of the Precision afforded your pride beliefs the way that you think the world works or the way that these particular sensory uh path of sensory stimulation are generated relative to the evidence at hand through that you're accessing through your Sensations so when people well suddenly when people like me and you talk about um this resistance to Illusions um in people's African schizophrenia what we are effectively saying is that they have less precise or imprecise prior beliefs that they're more sensitive to any sensory evidence at hand so if Illusions are the things that reveal our prior beliefs and people with schizophrenia are resistant to illusions that means they don't have very precise primer needs and that tells you an enormous amount about the the message person the Dynamics and the sense making in people who are resistant to Illusions um particularly in relation to how we think the brain encodes that Precision because you have to estimate the procession doesn't come for free it's very much like a statistician Computing the standard error which is the inverse or the component of precision it has to be evaluated and if you get that wrong you can produce your type one your type 2 errors all that false influencer underwrites hallucinations and delusions but in this instance um a resistance to a particular kind of um not a pseudo hallucination but a particular kind of illusory person I I think I'm glad we're touching on this because this leads me into my next question which which is surrounding illusionism as a theory of Consciousness when you think of an illusion as a misinterpretation of a sensory stimulus that does exist um and you think of the theory of illusionism Keith Frankish Daniel Dennett Susan Blackmore lots of people working and to an extent when you take into account certain things that someone like Anil sits is with perceptions being active hallucinations um you you sort of come to this conclusion that okay the sensory stimulus we're receiving sort of gives us an idea a posterior belief at some point that we have what is known as a conscious experience a phenomenal first person subjective experience sorry I'm going to head into philosophy at this point and then we'll try and get back into the Neuroscience but at the physics Etc so when you think of Dennis and all these philosophers who discuss illusionism as a theory of conscious the basic premise is that we we are are experiencing something uh however because let's say someone like you or Donald Hoffman who takes Fitness payoffs over reality the fact is we are never really perceiving what is actually out there we're just perceiving what is meant to allow us to survive long enough perhaps um to breed do whatever we have to do and in this case Consciousness believing we have some sort of an entity something similar to an eon Vettel um the fact that we believe this allows us to go through life and successfully reproduce and continue to thrive do you think this has a lot of value and do you think that these theories of has value using your theory of free energy principle and the Bayesian brain to back it up well in the way that you articulated them absolutely yes okay um do you have any rebuttals against that not whatsoever um no I I think you phrase up um in a compelling uh way and um it fits exactly with the view of uh a Bayesian brain and you know more generally the free energy um your control hallucination um is just an expression of the way that sensory Impressions somehow control our sense making on the inside and the degree to which we let them control depends upon the Precision that we afford um these sensory Impressions um by definition will never actually know what's going on on the outside so the whole point of inferring is you don't know the answer you're just left with the best belief the best probabilistic um guess as to what's going on out there there will never be any true access to the causes of that and in a you know sort of complementary fashion I will never know what you are thinking I can never know so I'm on the outside looking at you and so you know it all including the free energy principle in fact um as applied to your brain this is just a hypothesis it's just a belief you know there is a fundamental divide that in the physics application of the energy principle rests upon something called a Markov blanket but there's a a fundamental divide an informational barrier um or probably you know beyond information an existential barrier between the inside and the outside of any thinking thing and which means that the the question is that what is what is reality is there a metaphysics is is unanswerable um all you can do is um I think an incest uh expresses it as inference to the best prediction as opposed to explanation um but I think influence of the best explanation is perfectly okay in terms of explaining the world to yourself as you um sample it as as you send it um one thing um so I'm not a philosopher uh yeah I am very naive and and however um I did want to make a philosophical Point here um that we you know you use the phrase We believe uh in some aspect of ourselves including the fact that we believe that we have experiences there is an argument that this is just another illusion it's just another hypothesis that in fact selfhood in and of itself is just another hypothesis that these are sort of illusions that we've been to the table to make parsimonious sense of the fact that there is there appears to be some coherence in my embodied interaction with the world um but it's not a necessary belief for living and surviving and adapting and I'm sure if viruses could tell you they would deny having a sense of self um so I yeah one night to argue that Consciousness either the kind of self-consciousness that you refer to or just um the notion that I am having a positive experience you know if the kind of um experience that people like Thomas metzinger would associate with opacity or phenomenal uh opacity um these in and of themselves are still just hypotheses they're just ways of explaining my sense making in this instance on the inside as opposed to explaining what caused my external sensory states does that make sense but how this I think is entirely consistent with you know with the views that you you expressed I think it's it's sorry for you continue call well no you asked for a pushback pushback I can think of um and again forgive me if I'm I'm using the wrong words but the free energy principle inherits from um the theory of sparsely coupled random dynamical systems um and the special case that emerges when you have a particular kind of sparse coupling between something and everything else if that is true it does have if you like a metaphysical commitment that there has to be something else out there so um there is there is a sense that you can um you can be a complete skeptic and there is no reality um uh from the point of view of applying the free energy principle um to my sense making or indeed applying it to your sense making your sentience um but the free energy principle of itself starts from from a very metaphysical uh position that there isn't out outside and there isn't inside um and what we're looking at is quite simply a generalized synchronization between the Dynamics on the inside and the outside that can be read as inference and belief updated I think I think that's exactly where someone like Daniel Dennett would break away from someone like Donald Hoffman exactly what you're talking about where that it would suppose that there is is a reality there is an external reality however there's still no such thing as phenomenal experience or conscious experiences because we are crafting this as we're going along we're interpreting that way and that might not be the best way to say it anyway whereas Donald Hoffman then goes to the assumption that all conscious experience is the reality we're talking about and that might take it a bit further because now there is no external reality we're sort of creating reality from conscious experience but yeah he takes a different physical route uh just called briefly what I wanted to say was I mean I've got tons of notes here I just want to show you a lot of different things I've got a bunch of equations here the main goal was because I've seen so many different talks of yours and so many different interviews I've noticed how people tend to as I said ask you questions based on their expertise or their field of expertise and I didn't want that to happen because for the most part I'm so on board with your view and I love the theory so much that I used it in my own Master's dissertation to defend illusionism so I the dissertation is called psychiatrist defense or illusionism has a theory of Consciousness and a huge chunk of that actually Incorporated your work yourself call uh Chris frith and many others but I was worried that when I get into this interview I'm going to start defending it even more and I did not want that to happen so I made very specific notes to Sidetrack me off off that path but before I do leave that part the last thing I do want to say is if someone does ask you what Consciousness is how do you actually go about answering that question do you find that you go down it philosophical route or do you stick to a neuroscientific psychiatric Road yeah but I think very much like you um it would depend upon who I am talking to and who is listening uh it would be very context sensitive yeah there are but yeah if you're talking to people who are um in a more physicsy world and they they are interested in the in the Naturalization of Consciousness I would appeal to this um well go right back to the your uh Cartesian dualism how do you repair that or should you repair that any other there is actually an argument I've heard recently from very valued colleagues um that there is a neocartesia cartesianism um that emerges from specifically uh the free energy principle on the one hand and Quantum information theoretic formulations on the other hand um this is due to things like the inner screen hypothesis by Chris fields which is uh you know a fascinating resurrection of the Cartesian theater of a particular sort um but if I was talking to to somebody wanted to naturalize um naturalize in relation into the laws of physics writ as usually differential equations or in Quantum information theoretic terms I would be looking to answer their question in terms of how how do you how do you bridge these two um the you know how do you arrive at some kind of dual aspect modernism um that's still preserved The Duality but explain it um and I would be talking about the difference between a posterior belief a Bayesian belief that is encoded by the states of a brain for example and the probability distributions that are entailed by the states of those brains so you've got on the one hand um an intrinsic information geometry that inherits from the probability distributions that somebody in the thermodynamics as a physicist would understand the kind that we use in brain Imaging for example to look at brain activations through the expenditure of thermodynamic energy in order to do neuro you know to support neurodynamics about exactly the same by physical states are those that parameterize another kind of information geometry which is the information geometry of beliefs the space in which I do my belief updating moving one belief from this point to this point because that movement is important because most of the um much of the arguments that I would engage with in terms of Consciousness would um frame Consciousness as a process you know I think you mentioned that explicitly what does that mean from the point of view of trying to naturalize a conscious process that was quintessentially belief basis is movement on some statistical manifold so you do get that for free from the free energy principle so it's you know that's why I and a number of uh at least close friends and collaborators uh use the phrase you know a physics of sentience sometimes if you don't want to be quite so bold and sometimes as you get told off for using the word sentence so but at least you can talk about a physics of um and it is that sort of link between the information theoretic the belief the statistical um aspect of neurodynamics and the thermodynamics that underwrite that which does I think provide that link between um the the other men's well you know the Cartesian dual dual aspects um of um of certain things like the brain if I was talking to uh um people um in philosophy who are more interested in um you know phenomology I think you have to address some hard questions about you know what would be the necessary properties of a self-organizing self-evidence free energy minimizing system that could possibly support um qualitative experience or um qualia um as uh and then you have to ask other questions about um when would there be some self-awareness of this and um you know issues of self-modeling of the kind again um addressed by people like uh Thomas metzinger I about those kinds of um conversations would take a slightly different route and would focus on the phonology and interestingly they come back to the the Precision we were talking about before if you remember we were talking about um having to estimate the Precision which means that you've got well for those kinds of systems or creatures or particles and that have the capacity to estimate the Precision of their implicit beliefs and the sensory evidence that is updating those beliefs that the very Act of uh deploying those estimates of precision and inferring that Precision becomes a kind of mental action so now you've got a way of linking um mental action read as people like Jacob uh limouski and and Thomas metzinger and that kind of philosophizing um with the psychological reading of this mental action which is just attention it's just basically um deploying or affording more attention to this level of representation in a hierarchical context or this um sensory stream thereby selecting that for leap updating giving those sources of evidence privileged access to your um deeper higher level uh belief habitating so you now you you're in the world of Building Bridges between attention mental action and filmology and then you have to then you start to talk to people who like to think about self-hood and self modeling and self-awareness um and you start to think about well you know how do these things unpack in terms of the kind of generative models that are necessary to Define model evidence marginal likelihood variational free energy what kind of models would you need and you normally get to you know very very deep models uh hierarchical models where normally we have in mind a sort of hierarchy or a centrifugal hierarchy so you know deeper is is usually sort of more abstract and slower um and um you know one normally starts to think about well what kind of depth would be required for me to have a sense of self or first of all I have to be an agent to be an agent I have to have a generative model um that entertains the consequences of my action otherwise I can't plan so immediately we're talking about generative models entailed by certain creatures or people that escapes the moment simply now because I've got a journey model of the future unlike a thermostat or a virus I've now got a very deep generative model of a particular sword which goes into the future it's future pointing but crucially it's my future it's not your future anything else it's just my future what will happen if I do that so that makes me into an agent so immediately talking about a very special class of of self-organization self-evancing um and then you'll have to ask the question well at what point do I recognize that it is me that is actually doing the planning and and why you know you know yeah if you want to apply the theology principle you have to ask the question well what licenses this kind of Deep generative modeling um and that's a pressing question because the um you know I've used the phrase self-evidencing I should just very quickly um qualify that which is a philosophical term which I know nothing about in philosophy but I do know that Jacob Harry my friend in Melbourne is a philosopher uh thinks it's quite apt to use as a description of the the Bayesian brain minimizing it's free energy because if you remember the free energy is synonymous with technically a lower bound on surprise surprise itself information but it's also the negative model evidence or log model evidence so as you're doing um optimal inference you are implicitly trying to maximize the evidence for your models of your lived world so yourself evidencing um so the interesting twist here is that um you can you can easily show literally on the back of a matchbox that the the log evidence is equal to accuracy minus complexity which means that every move I make in every aspect of my brain that is engendered during neurodevelopment or indeed Evolution that's to pay a price for every more accurate explanation illusion um that um that it um deploys and that prices the complexity cost so it means you're trying to minimize the complexity so why on Earth would um say Evolution equip you and me with this really deep generative model where I've got all this extra complexity that I can be in this state of mind on that state of mind I can be anxious I can mean that I can be frightened I can be embarrassed and you know why would it do that well of course if it provides a parsimonious explanation of my sense making and my active engagement with the world then um it pays its way in terms of the um the the the increase in accuracy of the kind of things I can explain and engage with pays its way or over compensate sufficiently for the increased complexity and then you ask yourself what kind of Worlds would potentially um justify this kind of modeling this kind of very sophisticated hierarchy deep modeling um and usually one ends up and now I'm talking to people in neuroethology and social Neuroscience you end up with well the only kind of worlds in which I need to infer I am me and I am in this dispositional state is in a world in which there are lots of other things that are exactly like me that have certain dispositional intentional status in relation to me otherwise I don't need this it would be you know I'd have an overly complicated model so you argue yourself into a situation where this kind of Consciousness it can only be present or is only licensed under the free energy principle in a world full of con specifics and creatures and usually in an encountered eatonish that depends upon lots of other people so that now becomes a slightly relational aspect uh to this kind of Consciousness so I've given you a I haven't talked about the the the in a screen hypothesis but there is another quantum physics version of this so there are lots of different stories you can tell depending upon what people want to hear Carl you touched on so many different ones there I love the last one because it reminds me of Chris Fritz work and and he loves to talk about this social and relational aspect of Consciousness you guys have written many papers together and someone else you've written with is a colleague of mine from South Africa a very good friend Mark salms you guys have also written together and I know Mark takes us down to a more basic level of of existence in terms of conscious experiences coming from more basic parts of the brain what got you guys to write here then how did you guys sort of get together and do this these beautiful papers together right well I just mentioned both Chris and Mark in that um so interestingly um when I um as a young psychiatrist got into brain Imaging and it was at exactly at the same time that Chris was Chris fifth uh aim um from basically clinical uh psychology and neuropsychology that was one of the world's certainly the UK's experts the schizophrenia research he came to join us at the MRC Cyclone unit in the very first brain Imaging also in the mission tomography studies of people with schizophrenia so we you know he's been my mentor and friend for since the Inception of my academic career and and is still meant to be writing the last paragraph of the last paper that I've tried to submit at the moment uh Mark but um there's an interesting connection and um comparison between Mark and and Chris Mark I met later in life at International conferences and was very impressed with in those days his encyclopedic and um very embracing knowledge of Freudian ideas um in relation to psychoanalysis but also neuropsychology and so he's one of those polymaths um who can transcend different fields you know he can he can write volumes on Freud and yet uh bring the most informative and puzzling neuropsychological cases to the fore to events some very deep questions about you know how we experience our world and you know how that can be expressed via things like neuropsych analysis so um the interesting thing is that Chris really hates Freud but they're both they're both committed to understanding the brain as an organ as a social organ and uh but as you write this say uh Mark is much more on the sort of you know it predicates his work on the neuropsychology and the neurobiology um whereas Chris is much more in terms of the dadic and the social and social Neuroscience aspects um that underwrite um you know sense-making way it matters um uh but both of them at the end of the day I I try to understand that you know the patients they saw as young men and perhaps Mark is still seeing patients I don't know um so that that's how I got involved uh you know Mark's ideas yeah I think you're absolutely right they um they are um more grounded in the wet wear that um that would have this um that would endorse certain um psychoanalytic psychic therapeutic psych neuros and neuropsychological and psychiatric readings um and interestingly he's happened upon um exactly the same Focus that I suspect your computational Psychiatry thesis focused on so you know again we come back to this key notion of precision and uncertainty and of course you know if you want a you know to naturalize uh it's all about beliefs and if you want to what is definitive of a belief well it has clearly uh when thought of or when when read as a probability distribution it has an option an attribute of content in terms of the location parameters but more importantly it's got a an attribute of uncertainty a dispersion that's the whole point about a belief as opposed to a point estimate so if we are creatures that deal in beliefs then the you know the challenge that we face is basically evaluating and knowing and representing our own uncertainty either personally or some personally you know and indeed um you can see this throughout say in economics you know the markets have lost confidence there's in you know things that matter are usually those the confidence the uncertainty that I that I hold my beliefs with um so that that I I keep using the word uncertainty because there's a lovely little phrase that Mark has celebrated his most recent book that you know the Hidden Spring and subsequent sort of academic Publications which is this notion of felt uncertainty so he thinks that feelings uh all about the estimation of and deployment of decision and uncertainty in exactly the same way that we ended up trying to naturalize um resistance to Illusions in people with schizophrenia it's all the same thing so so he puts a very heavy weight on that um to the extent he gets very cross when people become cortical Centric you know they think it's all all in the frontal lobes there's no he says no he's literally deep in the hierarchy literally deep in the cells of origin of these really powerful neurotransmitter systems that do the attentional game they do the neuromodulation they set the Precision they balance and coordinate our our belief updating and of course that fits very comfortably with what we've talked about before which is a mental action read as attention being able to get some control over those um Dynamics and belief update businesses that basic mechanics um what's going on inside our head so when I mean you guys both of your papers I remember you you all sort of culminate and get together with this idea that this Bayesian brain this this is fundamental to what we do when you think about perceptive experiences we realize that at some point my red sunset and your red sunset are not the same um is a very two different experiences both comprised of different neural

Euro factors different social factors different experiences in the past so even time plays a big role this comes along well when you think of Markov blankets because you think of these different internal States um different active States perhaps for each person who's thinking how do you find Marcus blankets play let's first talk about how Marco blankets to play a central role in the free energy principle and how this sort of navigates across different areas from physics to chemistry to biology to psychology um let's go to economics Etc right I thought you were going to talk um synchronization across Markov blankets and communication to get to that at some point uh excuse me so um yeah as you say the the mark of blanket plays a foundational role in the physics reading of the 300 principle where you have to before you can start to talk about anything you have to be able to demarcate and Define what you mean by something um and the first move is um to individuate that thing from everything else how do you do that well you simply identify those states that intervene between everything else and the thing in question they separate the internal States Dynamics paths narratives technically um you know these would be States in a state space um from the point of view of say um a large van equation or um some kind of statistical physics um and the um what transpires is that you if you want to individuate the state's internal to something from the state's external to something then one way of doing that and one could argue the only way of doing that is to demonstrate that they are conditionally independent if you want to be able to separate there has to be some kind of probabilistic separability there has to be some kind of Independence but it's conditioned upon the Markov blankets so what does that mean well this basically means that um you and I or more specifically your brain and my brain um are not closed systems we are open systems and our systems are open to our Niche or our environment or our heat bath if you were a physicist um but that kind of openness is um the kind that preserves that conditional Independence between me on the inside and everything else on the outside and that's where the Markov blanket comes in and then you can um for most things partition the Markov blanket into two sets of states the active States and the sensory States and semantics are rather obvious in the sense that it's just making a distinction between those states that mediate the exposure of the inside to the outside so the influence the vicarious influences of the outside on the inside and those have been the sensory States and then to complete the circle to induce the necessary kind of circular causality we need for the synchronization we're going to talk about you have to have the other side of the street um which is the active states that mediate the influence of the internal states on the external States so you've got this rather delicate fundamental sparsity structure where the sparsity is read in terms of the influence that one state has an on another state that foregrounds a special set of states that are called blanket states that come um be divided into active and sensory states that intervene between the external internal States um the inner screen hypothesis we were talking about before inherits from a Quantum information theoretic formulation of this notion where you can look at this blanket as a screen onto which I can write say the environment or the world out there writes onto this blanket writes onto this holographic screen and those become my Sensations and I can then read those Sensations from the holographic screen on the inside and then I can double back so that the environment senses me through my actions so there's a if you like one level of perfect symmetry between the two and that's symmetry um is just um so that symmetry is advanced by um the minimization um of free energy in the sense that if I can predict what's going on out there and what's going on out there is at the same time in some sense trying to predict what's going on inside my head the ultimate the free energy minimizing solution the most likely solution just brings us back to you know being in characteristic States for the environment and the need but so we're talking just about a description of a generalized synchronization or synchronization of chaos um between two sparsely coupled dynamical systems with the special structure that allows me to preserve my conditional Independence so in a sense the uh the um The Illusionist perspective we were talking about before is a must it can't be any other way if you define existence as conditional independence from the world in which you are immersed but of course there has to be a world in which you are reverse so you can't take that too far in terms of a skeptical position yeah when you think of Markov blankets um in the context of let's say neural networks um how could the concept of Markov blankets be used to better understand flow of information or computations within a neural network is this applicable and are there any specific applications we could utilize this concept to enhance our understanding of Consciousness neural connections Etc yeah I think I think there are it's interesting that you should ask that and what you should do is get um Maxwell Randstad and Chris field and possibly even Mark I mean Mark Mark Sims has also been involved in these conversations and another other people I'm sure they will also suggest so this group have been thinking for a year or two and are currently submitting papers to things like the general Consciousness studies and elsewhere that rest explicitly on um halving the internal dynamics of a brain in terms of nested Markov blankets I'm trying to ask the question is there something unique about a Markov blanket on the inside in your brain which you know may be physically very distributed um but statistically well defined that could possibly support Consciousness so this is the inner screen hypothesis that I mentioned before so your question you and perhaps you've actually said seen a preview oh well it's a very it's a very impressive to astute question then because this is exactly what people are like Axel and Mark and Ken and uh David and everybody else would love Chris would have loved to have been asked um yeah so that's the whole premise that that well first of all it's this stand back um so you're right to ask yo are Markov blankets useful um yeah I think I think that they are foundational and they're used implicitly every everywhere I mean not only are they foundational in the sense that they um they Define Your Existence and my existence and anything that we uh that we engage with or we think we engage with um but they are implicitly you used whenever you start to simulate or reproduce or emulate any kind of sense making or intelligence artificial or natural um you know whether you're playing with neurons neural cultures in a dish they are the cell membrane is a Markov blanket that you know defines a cell the connectivity of the neurons defines another scale of Marco blanket or a marker plug into another scale which says that this is a network uh not exactly the same maths applies to neural networks and machine learning and artificial intelligence research and the operation of your iPhone or Google search machines um so these things become practically really important in defining the network of a thing that's doing the um the you know the computations of the information processing in a pragmatic sense and indeed you can read these active and sensory States as the inputs and outputs to anything that is doing some kind kind of computing um so they are essential um in defining things like say one human architectures you know we do not you know you and I don't use a value but not connection but you can still use knock-off blankets to Define them but even if you look at the kind of uh architectures um that we use um for sense making in the hierarchical Bayesian brain or hierarchical predictive coding um story then the message passing that is required to do the belief updating becomes tremendously efficient due to the presence of these and carefully self-maintained self-evidencing Markov blanket structures so remember the mark of blanket rests upon a sparsity structure it's what the connections and the influences that are not there so that means that you can now do local computations if you can Define the Markov blanket of you uh you may be just an element of your dendrite dendritic tree you may be a synapse you may be an entire dendritic tree you may be a cell you may be a brain region you may be Compass you may be an entire brain you may be having a type person or you may be a family you may be a species you may be an institution you may be a biosphere you know and every every scale the same rules apply um whereby you get the efficiency from um the sparsity where the sparsity underwrites the very structure and the ability to actually dissect at every scale and the system at hand so that leads to an IEP that's practically really important when you're designing neural networks or chips you you need to keep all your computations as local as possible and you do that by using your Markov blanket um if you wanted to if we return to um the implications for um belief updating in a brain um then just saying the phrase hierarchical predictive processing or hierarchical basic inference immediately commits you to a certain class of Markov blankets because the hierarchy is just defined in terms of the connections that are not there so each hierarch level possesses a Markov blanket that separates its superordinate level from its subordinate level so just talking about hierarchical structures is just one special case of a particular kind of nested Markov blanket and then we get to is there a special Markov blanket so now this is the Crux of the paper that you have read that I thought you had I'm looking forward well I'll send you well I'll ask you know if you email Maxwell ramstead and Montreal he'll send you um he'll send you I've only got a copy with all the authors redacted so I'm not quite sure who the authors are at the moment um I'll take up that option I'll take up that offer and definitely do that no he'd be delighted um so the other the idea of um one of the key ideas um that fits very comfortably with um the ideas from Chris field and people like Jim glazebrook and Mike Levin um which is this inner screen hypothesis is that there is one irreducible Markov blanket there is one that you can't get inside and find any hierarchical structure or any partition what would that mean if there was one or more possibly but there was certainly at least one irreducible Markov blanket in my brain it would have some quite profound implications um it would mean that it could never see itself other than acting on the outside which is the rest of the brain so the only way you can solve evidence is by engaging in mental action and now you ask well what kind of mental action would it engage um would it uh what did the executive systems the active states of this particular minimal Markov blanket they are precisely the cells of origin of the neurotransmitters that Implement Precision control and attention so we're right back with Mark Sims again but we're in a very comfortable position because not only are we friends with marked zones but we're also friends with everybody who talks about things like higher order thought Theory so predicate the entire argument on this hierarchical structure which means that at some level and usually it's the higher level or the deeper level uh don't do interesting metacognition uh things start to happen that you could associate with uh with conscious processing but this is exactly what the minimal Markov blanket uh irreducible Markov blanket would say it is something that is deep within a hierarchy that cannot there is no higher or deeper level because if there was you'd be able to partition the internal States within that minimum Markov blanket it also fits very comfortably things like the global neural workspace

Theory I mean what is global and what is a workspace well it's something where you can share stuff so you can't further subdivide it and it's firmer because it sees everything so it has it has a lot of construct validity in relation to a game where we started everybody's right but they just got their own way of expressing it um but in this instance we are or one is expressing it explicitly in terms of individuation separability read through the lens of the free energy principle in terms of Markov blankets crucially in a scale applied a particular scale and this scale is the scale of brains so this argument wouldn't work when you're trying to describe the weather or Evolution it wouldn't just work when you're trying to describe things at a Quantum level or indeed um at the level of a um your single neuron or neural population this kind of argument is scale is tied to a particular scale but it has if you like a scale free aspect in the hierarchical uh in the hierarchical partitioning that induces the notion of a non-partitionable or irreducible Markov blanket so funnily enough that that notion of the irreducible Markov blanket being the inner screen is exactly the new Cartesian theater that we're talking about that's very fascinating I mean I'm scheduled to chat to Mike living soon and I had no idea uh I think I should send him an email to Chad about this you said it was Mike Levin Mark storms Chris Fields who else was there was one more name you mentioned the theaters the inner screen hypothesis that um if you like survives um a Cornerstone for this more Collective inclusive effort to try and get you know what value of I should call it a minimize a minimal unified model of Consciousness that includes people like Mark zones and and Kenneth and Wilton and uh David Rudolph and lots of people yeah sort of organized by Maxwell ramstead um and Chris Fields is a really important player in that but the inner screen hypothesis in and of itself um does indeed um uh is is indeed due to um uh Chris fields and his friends and particularly Michael Evan and people like uh Jim glazebrook so that that that formulation is another if you like access which I you know you'd have some great conversations with both Chris and Mike um so they come at this from the point of view of um sort of basal cognition the notion that everything is effectively um can be cast and possibly even is just a process of information you know is this information processing and self-organization just is the uh the processes that evolve that effectively do believe updating or process information using classical um you know when it comes to sort of the kind of biotic self organization that might Levin deals with so he you know he's famous for Zen inbox and all sorts of really insightful takes new takes on um cellular and um and your biological self-organization um you know some of my favorite ideas are you know if you'd that you might like is that you know when when you appreciate that sort of you know um physics and this is the Chris Fields quote and there is no bright line between sort of physics biology and psychology they're all the same things um but just using different words and and Mike takes this to his extreme or used to is is got more sophisticated uh in the past few years but uh several years ago he had this notion that um cancer was a was a delusion a delusion that arose from a pathology of basal cognition which I think is what you know cells are failing to infer know their place their failing to infer that no I shouldn't be growing in this context because it's delusional I should keep on growing and of course he's absolutely right uh so you can even get a computational Psychiatry of cancer out of this but it does require a commitment to think about all self-organization as basically a process that have made red as an inference presence under a belief that basic dating process the right kind of processing and bounding self-information and and entropy read from the point of view of information Theory Chris fields's particular take comes from uh his skills with Jim glasbrook in Quantum information Theory so that's that's where the information theoretic aspects come and then you can start to talk about the relationship between the principles of unitarity and in tanglement and the like and what that means when read um or what it means in terms of a Quantum formulation of the classical reading of the free energy principle that doesn't normally deal with Quantum you know it usually starts with uh you know classical mechanics with all the defender formulation I'm like I just I love so much about what you just said because we when I talk about it I often think about the same thing I often call ourselves receiving persons organized by organelles crafted by cells designed by DNA manufactured by molecules assembled by atoms forged by Fusion over forged by Fusion via Stella supernovae so this is a saying because I think if you don't incorporate all these things together you're just you're missing out an opportunity not to reduce things to the nothing but aspects but rather to grow the complexity of what we're talking about you just increase the scope of the conversation so much more by discussing every realm of reality and I think you get that and you seem to be working on that from that perspective as well absolutely and that that again I think that's a yeah I love the way that you actually expressed it you should definitely get Chris Fields uh on on your podcast because his big thing is that scale freeness that's scaling variance um that you know and in fact I was on a call literally um about two hours ago where he made um I thought a delightful point that um because the free energy principle is by construction scale free it cannot be reductionist at the same time so it is quintessentially so what you just expressed I think covered all those scales um and I think that's part of the well saying it has to be it has your your formalism has to be skill free it has to be devoid um for me mathematically in the spirit of the renormalization group it has to be deployed time and time and time again um and as soon as you realize that then you start to answer the D or ask D questions about you know top down causation between scales so as soon as you've got a scale free Theory then it can apply either to camera oscillations in your hippocampus or it can apply to Natural Selection red as Bayesian model selection and then you start to ask big questions well how does this very slow self-evidising process contextualize and be informed by this very fast self-evidising process or information processing in a way that is emerging from just having these same kinds of processes unfolding uh with a separation in this instance temporal scales so you know Chris would wax lyrical for a long time about this so perhaps you can get Chris and Mike together then you I think that's a real idea I think I should start doing that I know I know the last time I think you Kurt and Mike I think spoke together as well if I recall correctly I can't remember yeah yeah because I wanted to ask you when you think about this project that everyone's talking about um and Markov blankets do you think that because some systems are so complex how do you find do you think there will be any sort of indirect or hidden causal relationships that might hinder this from happening what what sort of applications or techniques should be employed to effectively identify Markov blankets within complex systems um yeah so this is a very sort of practical uh uh issue um and it would be very nice to be able to sort of parcellate the brain for example or parcelate um an institution or a you know uh you know an economy um to be able to identify the uh the the different Markov blankets um given some empirical data um there is a worked example of this which is little known because it's just so complicated uh it's almost impossible to read but I wrote it and I'm quite proud of it um and it it I think it was in uh Network science but it was exactly the challenge of taking brain Imaging data um thousands hundreds of thousands of measurements of activity over time and then parcelating it at different scales so it was exactly what we're just talking about it was creating Markov blankets of Markov blankets and Markov blankets I'm doing it in a way that was empirically uh constrained by measuring the connectivity and it you know at its sparsity to imply where the market blankets were and then once they've got a Markov blanket those now constitute the Intel states with a bigger Mark of magnet and then you can sort of carve up the entire brain successively course from Corsa um spatial scales and what was really interesting as a phenomena that um is if you um I was going to say immersion but in fact it's actually baked into the you know the uh the renormalization group that you would access mathematically um that we would use to access the scale-free behavior at least formalize our naturalize it um that things get slower as they get bigger so you could actually we could actually estimate the time constants of the neural fluctuations and implicit belief updating associated with different spatial scales ranging from a cubic centimeter or millimeter or so of a brain through the entire frontal lobes and then we can extrapolate out to very very big things and very very small things now look at the characteristic time scales it was just implicit by scanning through the time scales by having lots and lots you know literally a hundreds of thousands well at least a thousand uh a thousand little Markov blankets all jostling together uh partitioned uh and they're moving to say a hundred of them and then to ten and it's by extrapolating you can then think about sort of collections of brains pen brains or a tenth of a um a cubic centimeter of the brain right the way down to to the quantum scale and let's look at the different time constants that they were engaged so that was a really interesting academic exercise but did rest as the same being able to identify uh Markov blankets um what you're asking though um is you know is is a key issue in terms of complex system um analysis and modeling uh being able to carve the system in the right kind of way um in order to um usually to simulate it in a way that allows you to do forecasting or scenario modeling and the like or uh projecting uh a response to a therapeutic intervention and say a clinical Neuroscience or Psychiatry through to weather forecasting yeah getting the right course grading the right Markov blankets in plain is absolutely essential there are lots of people working on that at the moment one email exchange event this week is from a gentleman called Jeff Beck who's taken on the very deep challenge of considering Markov blankets that wander around so one interesting aspect of a simple definition of a Markov blanket is that it depends upon um a probability distribution and that exists over a non-trivial amount of time but if you think of something like a candle flame that is moving um and furthermore the constituents the molecules and the ions that constitute the plasma that is the surface of the the flame is it fair to say that the flame has a Markov blanket given all its contest constituent molecules I.E other small scale markup blankets and continually being exchanged and um moving wandering around so he's taken on the challenge of of taking Markov blankets into this sort of um more liberal domain and dynamic domain and see if one can still apply the free energy principle in in a practical way but to do that he has to have at least simulations or numerical studies that allow him to actually identify Markov blankets that move like waves in certain uh in certain media so that's still I think an outstanding problem which a lot of people well when I say a lot a few key post callings looking at and I think it's a very useful company of inquiry that's quite fascinating it's great to see that so many different people are working on this at this point because when you look online and you look at the amount of citations I mean you are one of the most cited neuroscientists on the planet and it's great to see that a lot of people are taking these Concepts seriously and trying to roll with it looking ahead to the Future what directions or advancements do you anticipate in the study of Markov blankets for example except for the one you just mentioned where do you think the free energy principle is going from here um I think there are probably three directions of travel um and we've implicitly we've already covered the most important ones in our conversation and so you know the first one is indeed computational Psychiatry and computational uh neurology I mean in a sense all um the work that Christopher and I and many other colleagues put in um you know decades ago into understanding functional brain architectures via brain Imaging was in the service of your ultimately asking questions about neurology and Psychiatry and much of the theorizing that led to active inference and the free energy principle inherited from the data analytic and modeling that we were applying to uh to make sense of brain Imaging data I realized at exactly the same principles could be applied to the brain on when I say realize we're told by people like Jeffrey Hinton you can apply exactly the same principles to the brain the health helps machine um so um that's one that's one direction of travel I think it's just uh again I use the word naturalizing and I recently learned what it means and I think it's you know a useful phrase to to naturalize Psychopathology in a way that you can write down the laws and the maths and you can start to creating silico patients and digital Twins and intervene on them uh without actually um you know giving drugs on uh psychotherapeutic interventions to patients until you've shown um and to a certain extent that that style of approach is um already starting to bite not under in the context of the free energy principle explicitly but certainly um in the modeling of things like epilepsy you know have it having having naturalized pathophysiology say you know the transmission of epileptic discharges and seizure activity in the brain uh to the extent you can actually put it in a computer and start to do weather forecasting of this particular person's brain and what might happen if we did this and that I think that's one very exciting Way Forward uh another complement of that um sort of Applied application of the free energy principle um would be the ability to phenotype people so there's a phrase computational phenotyping which which I quite like which basically means being able to characterize this person or this cohort um in terms of the parameters of their generative models that they use to explain their world and that provides a really powerful and efficient summary on this person and the in terms of their beliefs and their belief updating and their prior commitments um and being able to to be able to measure and quantify somebody in that kind of way can be very very useful not only in terms of tracking responses to therapy but also just getting the right there's allergies the right way of carving up various psychiatric conditions and their etiologists because now you've got a naturalized process Theory under the hood so that's one kind of thing the other direction of travel I think is in artificial intelligence um just actually building so practically what the 300 principle brings to the um is not really a pretty first principles account of uh existence and sentient behavior um but practically another strange inversion another strange inversion um instead of having to um instead of having to understand the system in terms of its Dynamics as you would the complex system modeling um you're trying to understand the underlying mechanics by saying well these Dynamics this set of say differential equations looks as if it's performing a gradient flow on this lipunov functional this uh lagrangian and I'm going to try and estimate what this um what this underlying potential function is that causes my Dynamics and a simple explanation for the complex system I want to understand um for me of course that's always going to be this surprisal or the the uh the log marginal likelihood or the free energy balance on that but what you can do is if you know that the the Dynamics of your complex system and always be expressed as a gradient flow on a free energy functional objective model where the gender model just scores or articulates characteristic states of being that you want to um you want to emulate then what you can do is you can just write down the free energy functional of your preferred states of being and then just integrate the system to reproduce the sentient behavior that you would have seen if this thing existed and had these was attracted to these characteristic States so it's a really powerful way of building machines that look as if they exist in a particular way and of course that's the aspiration of machine learning and artificial intelligence you um and so it's only in principle you could deploy the 300 principle in the context of building artifacts that do the right kind of belief updating by the right kind of course I mean uh being able to model their world where their world is it's basically the world of users basically you and me which brings us to the third direction of travel which is um the kind that Chris fifth would like which is you know the explicit acknowledgment that what is really interesting here is this scale free application of 300 principle where you've got lots of different free energy minimizing self-evidencing Markov blankets all living in Markov blankets over a different scale in institutions and theologies and now ideologies and uh you know with ethnic bounds and and all that so how does that unfold and how do you get the right kinds of ecosystems that have this joint free energy minimizing behaviors that does not um incur upon or require certain Markov blankets to be vitiated or destroyed you know so what what we're talking about now is the physics of an ecology that has sustainability at its heart or at its core because the whole point of the free energy principle it's a description of systems that have an attracting set and how they sustain themselves within that attracting said so I would imagine that the 300 principle will be useful and indeed pursuing that through colleagues that and versus AI uh so my colleagues in Industry um this this is one application the free energy principle in the service of um in you know underwriting in a principled way the aspirations of all right-minded people engaged in designing the next version of the world the World Wide Web or artificial intelligence or Information Services and speak not to growth that speak to sustainability that you know a biomimetic take on sustainable ecosystems but in this instance ecosystems of intelligence so that was a bit hand wavy but it's yeah it really is a fascinating game when you start to talk to people who are actually in charge of engineering uh the next the next web there really is so much potential um there's something I wanted to ask and I forgot about it earlier but I was thinking about Markov blankets and how they they played this role um to sort of just differentiate between let's say it can sort of play a role between differentiating between the self and another in a sense does it have any role or impact when it comes to free will how to incorporate these ideas together it might sound good on your side good continue with your audio no no sorry I was having issues I couldn't hear you hold me now it's back it's back hey sorry all your questions okay so when you think of Mark of the United States but irrespective and sequence uh foreign foreign foreign I think that sort of brings me to something I wanted to ask you from the very beginning with the free energy principle in mind what do you call is the meaning or purpose of life as well that doesn't really have too much of a that question has no meaning uh so so from the point of view the participants observing the universe that was comprised of things what will not the free energy principle is saying is that if this universe is um settling onto an attracting set and if they're all like a tractor then it must be the case that it is um on average maximizing the GMB attractive characteristic States it's just like rolling to the bottom of a bowl that's it um what way of reading that road into the bottom of the goal that convergence um the active set um is in terms of a minimization of thermodynamic free energy the special thing that's brought to the toe by the 300 principle is the insertion of the markup blanket now you can read the minimization of further environmental energy in terms of a minimization of certain beliefs inputs where the beliefs are held by internal States but they are about external States so this is before I I phasing like that because of your purpose so yeah that's the energy principle gives you a um what people like loves to custom and Maxwell Randstad would require a basic mechanics which is exactly the same as simple mechanics exactly the same as statistical mechanics exactly the same spot mechanics so this particular physical state has a meaning in the nation represents it's expanding it's a hypothesis about it is a lot of this external State about something else this illusion illusion of what might be out there so I think that he has if there is life in the sense of that thank you there is an emergent meaning and that means found in the internal dynamics of life and it's meaning uh is established in a relational sense of The Simpsons uh the um the relation um of the internal sense making um the measurement if you know Quantum physicist or whatever is causing things divisions and Sensations that you are using to do your measurement so this is a positive information Theory which I'm completely subscribed to and um so that would be one way of doing it I think it's slightly more um obviously popular as the question would be well what's the what's the meaning in my life where I am on this Julia systems that has its capacity or agency in the sense that I have precise Dynamics and I have the ability to model the consequences of my action in this very particular case where I can now infer what I am going to do in the future whereas something called expected for reality so you can you can compute the probability of a particular narrative action path uh trajectory into the future as um basically a any soft Max uh function obviously called expected theology so the path to the Future that has the lowest expected for the energy is the most likely that I will follow that will select very well I will select the path that has the lowest expected of the energy and what does that mean an expected for energy in the same way that you can Express log evidence as equal to accuracy minus complexity you can express the expected free energy as the expected information being minus the expected cost where the cost is a surprise that negative self-evidence has caused the a um uh a characteristic state so if you know they'll get other characters feel unbreakable and very poor dead dissipating um then that would be very surprisingly highly costly so you've now got two like affordances or imperatives or patterns that you can select um which is um which is why what I have reading at the meaning of life to be you know what it's my likely think what it's purpose what is my purpose well it is the um it is the imperatives are underlying the things I choose to do what do I choose to do I choose those things that minimize my expectations I will try to choose those things and maximize the expectation again or minimizing the expected cost the expression information going it's just curiosity so the meaning of life for me assuming that I'm avoiding all those constant states of being myself in danger of the world very much it's such a problem with my wife um that the majority of behind me is the Curiosity it's it's basically uh be a scientist yeah yeah I think I think that's a that's a beautiful way to put it Carl um there's there's so many things I want to ask you about but something I often love to us my guess is and I think it's because this gives us an idea of how you were crafted as a thinker and how you really developed this I think over time if you had to give us a Mount Rushmore of your favorite philosophers or scientists who comes to mind instantly who are these people who really guided your thoughts growing up and made you the manual today because I'm pretty sure when people look at your work today you from the time when time moves on we're going to be talking about you have a feeling the same way we talk about some of the greats and I think it would be great to know who inspires Carl Preston right yeah I think that's a great option uh but what I think sort of the two people um when I read or reference with with warm feelings and a tinge of excitement um especially when I find their quotes um uh our um helmholtz and Feynman um Herman Von helmhansen and Richard Simon Simon um look the helmets reference that makes sense I mean when you think of unconscious inference and all your work it makes a lot of sense what about these guys really get you going well you're going about people like Mark Sims and possibly me um I mean he was a clear polymath I mean you know if you just think about every move you make on certainly every move I make in my little world um in trying to naturalize psychology and psychopathology and and more generally sort of you know what makes us tick and um you've come across some foundational idea or principle or decomposition that he is brought to the table so you've mentioned the notion of unconscious inference I mean that's just basically uh debating brain uh as beautifully articulated by um Peter diam and Jeffrey hinton's helmets machine in in the um the the early 90s um so that's certainly one thing but it doesn't stop there uh yo you you look at his work uh you know in perception generally you know he had all the one aspect of an illusion uh illusionists thinking at least and anything that was worth saying you know at some point he even said it but more than that of course you know um you know the helmholtz notion of um free energy and the contribution to thermodynamics is also pretty Central to what we're talking when I'm the kinds of things I talk about interestingly right down to something called the helmholtz decomposition which I suspect you probably won't know about if you're more interested in computational psychotic so this is a this is um an expression of the fundamental um theorem of um um what is it back to calculator no variation I can't remember my part is anyway the helmholstery composition just means that what you can do is describe any Dynamics in terms of two effectively orthogonal Parts one part is dissipative and it climbs Hills or climbs downhills like the ball rolling down the hill um this is the thing that generates energy it's a thing that does basically palpitating in an information theoretical free energy principle uh context um uh and the other part is called the conservative part also in a solenoidal flow and this part if you like circulates upon ISO Contours of potentials or probability distributions so this is if you like uh can be read in many many different ways if you're an engineer it's a decomposition of um procedures that you would use to assimilate data or estimate unobservable state to do inference as an engineer and using say a kelman filter and you would have a prediction so given my beliefs about states that what are our private section about what I'm going to sample next and that will be the conservative solenoidal part and then you would have your update which is basically a prediction error awaited by its Precision which changes my prediction my Prime relief um and that's the if you like the uh the dissipative part that's my engagement in the world that's that was that's what takes energy and does the belief updating but that is written down mathematically very beautifully by the helmets decomposition and what it tells is that if you've got this if um by appeal to the helmhouse decomposition you've got this solenoidal flow then you are necessarily breaking something called detail balance of the kind that you see in open systems and non-equilibrium systems so there's a direct mathematical mapping between the helmet's decomposition and the class of systems that physicists use and study which largely in the past few centuries has been sort of um systems closed systems at equilibrium in this Century it's now all about non-equilibrium systems and that non-equilibria aspect is just due to the conservative part of the helmets decomposition also it is exactly the thing that makes Us lifelike it's what equips us with life cycles with biorhythms with oscillations in the brain at every scale of gamma oscillations and our favorite hippocampal dendrite through to um your Beating Heart and respiration through to our sleep wake diurnal Cycles through to our life cycles through to the orbit of the you know of the heavenly bodies all of these that's not quite a biological uh aspect certainly for for biological systems that that break detailed paths and have this particular kind of itinerancy while still um maintaining their integrity and a free energy principle um then you've got a description of biotic self-organization the foregrounds oscillations and rhythms and recurrences returning to the states I was once in which of course is you know part of this sort of keeping yourself within within characteristic States so I've gone on a little bit there about how I know you can you can see why you said you're such a fan of helmets I mean it's clear you can see the joy in your face when you're talking about this and it's very fascinating work as well so it makes sense why this really gets you going and Feynman what about Feynman really because Feynman I'm a huge fan of Finance work um sometimes even for fun I end up watching fireman lectures just just keep home at night yeah yeah so what yes it's the one side so detainment that that I can enjoy my wife watches as well my wife is a nurse and not into into the physics uh but yeah no he's he's a very alluring personality um but you know if you didn't know this um it's probably worthwhile just saying out loud I mean the variational free energy was his invention um so he was trying to um well one reading of the history is that it was his invention you could have good arguments with um Russian mathematicians that can get at compression and um free energy minimization from the point of view of algorithmic complexity via things like uh soft uh induction and the like um and certainly there is a school of thought that really celebrates that sort of um compression view of um that is described by uh free energy minimization but the final view is is much more um more closer to my academic training and inheritance um so what he was trying to do was to solve exactly The Impossible marginalization problem that people would try to solve if they wanted to engage in exact base and inferencing exact sense making the exact statistics to actually get a grip of the um the the very best explanation of this metaphysical world that of course doesn't exist um but that's intractable in the same sense that he couldn't possibly integrate over all the paths of small particles say an electron would taken Quantum electrodynamics um he couldn't possibly um normalize that probability distribution by integrating over all the universe you know of paths you know a an electron could take so what he did was to dissolve that problem by converting this inference marginalization integration problem into an optimization problem and they did it by creating this band on the quantity he wanted to optimize namely the margin the log marginal likelihood which is the variation free energy so that's that that you know all of the free energy principle owes a you know uh um its greatest debt I think that's not quite Fair on help health and everybody else right through from account through to uh all the way through well everybody yeah I'm sorry people you couldn't mention but anyway certainly from a technical point of view um that you know that that that that pathological formulation is so beautiful um uh and EO it is leverage directly certainly in my personal work and the way that I would formalize the free energy principles I'd be elevated to you know you could read the free energy principles as duel to James's maximum entropy principle uh but in the you know when read or writ as a uh yeah in its pathological form that is really a homage to firemen's original conceptions but as you say it's such a delightful person to watch as well yes and and like me it was a committed smoker so I like that yes you love your cigars I know that the uh I often think about that video of Feynman um where he's talking about a red flower I think it was and and he says his friend his artist friend looks at this looks at this flower and then tells him you as a scientist you're obviously going to break this down and ruin it and he's like no no it's actually because I'm a scientist I'm going to look at the cells beneath the flowers I'm going to look at every single flow of each atom he's going to look at every single aspect of this flower and so much greater detail that it only increases in its its complexity and beauty and to him the flowers far more beautiful because of this and that goes back to what we were talking about earlier and I think that's something that you guys have in common it's beautiful because fireman has that perfect way of blending the science with the art and that brings me to this question I wanted to ask you when are you going to make or write a book because your work is so complex to a point where most people really don't know how to approach it when it comes to whether it's a neuroscientist psychologist physicist if they don't understand the work they're going to find it very difficult have you ever thought of writing a book that is a bit more for the Layman out there yeah uh well um no I I'm notoriously bad at writing um accessible um those horrible kind I think would be required for uh for a book and so I'm probably not going to uh what I've found tends to work best um is to um have the next generation of people who are much more fluent usually either um because they're sort of in a a person-facing kind of uh um science like psychology or medicine um or they've had a philosophical training so I find these these people are much better than I am and articulating in prose in narrative form and the ideas so they can write the Books I'll just sit back and buy I did try once um so foreign Robert Pryor MIT press who um has been uh asking me for at least three decades uh when I'm going to write the free energy principle book and he thinks he's got precedence because his surname is prior uh so so he thinks any book that appeals to the page especially the free energy principle has to be commissioned by him and he was absolutely right but the problem is there is stuff I can write is so uh talk to us and I have to say that you know as with the free energy principle itself with its emphasis on complexity minimization only providing a parsimonious explanation the 300 principle becomes simpler and simpler the longer it marinates certainly in my head so I you know I've taken now every year now I write a paper um you know describing how a simpler version and a simpler version that process will go on at some point it may be sufficiently simple to be attractive to Bob Pryor or any other editor who invites me or asks me the question that you've just been asking my suspicion is uh I will probably be retired or die before before that arrives so um you know but there are some wonderful uh Renditions of the free energy pro and con um and certainly uh most recently the active inference book by Thomas Parham Giovanna pazulo uh you know I actually say it there that you know I was allowed to write the forward but nothing else um yeah that's okay I was just about to ask you what are some of the reading recommendations I think that's definitely one that's on top that's actually a book that I want to read desperately um and I cannot wait if you had to recommend us any other books what type of books would you recommend to either increase our knowledge with regards to the free energy principle or active inference in general or just books here that you find help a scientist philosopher in general to grow in this field I'm afraid I'm not going to give you a very good answer to that there's a there's a reason for that there are two reasons um one I don't have time to read books um as when you get to my age um you you you you you tend to be you get asked to write blurb for books uh so I have to speed read certain chapters of books but I don't actually have time in my um working life to read books I can only review papers and and scientific proposals that's even working seven days a week any papers that come to mind oh there are lots of papers yeah but I I'm not going to say I'm not going to tell you what they are because they're the people I don't mention will be very very upset um so I'm I'm not very good at sort of reading books nor recommending books so what I'm going to do is is um what my but my father is so he sometimes sends me books to read which which he thinks are good for me so I'm just going to mention the last one that I read literally in a couple of weeks ago which is algo land yeah I really like that um so that would I would certainly recommend that and I think that had kind of written written that um last year as opposed to I think 2014. there would have been more emphasis because the free energy is just that you know that Viewpoint I think articulated slightly more in a classical setting in a particular pathological formulation of Richard Feynman um so I really enjoyed that but otherwise um if people are generally said there are lots of wonderful papers usually by philosophers usually by again people like Andy Clark and Jacob Howie and and Maxwell Randstad um uh you know when he's wearing his um his philosophy at um but there are also lots of really wonderful other but you just go to a repository so Baron has got a nice repository in GitHub of his favorite MEP papers there's something called the active inference Institute run by Daniel Friedman uh that is dedicated to trying to organize the material in a didactic or pedological uh way they're actually writing now I think another active inference book which is an explainer for Thomas's and Giovanni's actually in principle so if you're interested I think you go to the people who actually know how to teach and uh and and articulate these ideas it's funny because one of the last people well this was actually a year ago now that I think about it who talks about hierarchical Bayesian brains and active info it was Lisa Feldman oh yes yes and one of the books yes exactly and the book she recommended was alkaline and this was just when the book came out so it's funny that you both basically talk around the same topic and recommend the same book so it clearly is an impactful book in general well she's also written some very influential stuff in the emotional sphere and as now um you know so quite quite a key player in the active inference that's certainly the emotional inference and intercepted at princefield along with anneal Seth and and colleagues yeah just a close-up call um any theories of Consciousness the brain that you feel sort of go against the free energy principle anything that you know stands out that counter argues this principle in a way that does sort of get you get your Shield up um no I'm afraid not but I think that probably affects the fact that I don't have a philosophical training so I don't you know I'm not used to adversarial progression uh adversarial argumentative approaches um no I I don't you you know the whole point of the free energy principle is it has to explain everything or it explains nothing um so if there is a viable Theory Of Consciousness out there there has to be some way of if you like reframing it or um substantiating That Right theory uh by naturalizing it using the free energy principle remember Predator principle can never be wrong so it doesn't need to defend itself it's just a method or a tool very much like I'm trans principle inherently unfalsifiable yeah yeah and I think um just I'd like to say thank you so much for all your work I clearly supported in the fact that I I quoted you a couple times referenced you numerous times um so to me it is replicable it is work that a lot of us are using out there so from my side I just want to say thank you so much I call it it's a pleasure it's an absolute honor to actually chat to you and yeah I really enjoyed this conversation well so am I you've asked some really excellent questions you've kept me on my toes but also kept me smiling thank you very much